

Intergenerational Housing Allocation and Fertility

Part I: Compact Analytical Model

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*When does limited access to family-sized housing reduce fertility,
and which policies relax that limit?*

Overview

Model. Young potential parents decide whether to enter a housing market and choose tenure, housing, and completed fertility; children require goods (χ_i) and housing space (κ) per child. Old incumbents choose how much owner housing to keep. Rental and owner segments offer different size menus ($\bar{h}^R < \bar{h}^O$); buying requires a down payment and payment capacity; segment prices clear aggregate markets with construction.

Three results.

1. All access frictions collapse into one effective family-housing cap H_i^m per household. A binding cap raises the price of a child through the space the child requires.
2. Both markets clear, yet the allocation is inefficient when two measurable gaps are open: constrained young owners value space at $q^O + \zeta_i^{O,F}$, tax-protected old incumbents at $q^O - L_j^\tau$.
3. The fertility effect of a policy is a transparent local formula: exposure $\times$ pass-through to the cap $\times$ fertility response, plus entry, plus a tenure-switching margin whose sign depends on which tenure is the expensive way to occupy space.

Young households

Type $i = (y_i, a_i, B_i) \sim G_Y$: income, liquid wealth, inheritance exposure. Wealth is pledgeable only,

$$A_i(x) = a_i + xB_i + T_i^b(x), \quad x = 1 - \tau^b,$$

backing down payments but not entering the flow budget.

Children require goods and space; parents consume the residual:

$$c_i = C_i - \chi_i n_i, \quad s_i = h_i - \kappa n_i, \quad u_i^Y = \log c_i + \alpha \log s_i + \beta \log n_i.$$

Substituting $h = s + \kappa n$ into a branch budget with user cost q^m :

$$c_i + q^m s_i + \underbrace{(\chi_i + \kappa q^m)}_{\text{full price of a child}} n_i = y_i.$$

Unconstrained, a child costs its goods plus its space at the *market* price. Under binding constraints the space term will carry a premium — that is the transmission channel. ($\beta \log n_i$ rules out $n_i = 0$: intensive margin only.)

Tenure branches and the effective family-housing cap

Conditional on entry, households solve a renter and an owner branch and take the better one, $W_i^Y = \max\{W_i^R, W_i^O\}$:

$$W_i^m = \max_{c, h, n} u_i^Y(c, h, n) \quad \text{s.t.} \quad c + q^m h + \chi_i n = y_i, \quad h \leq H_i^m,$$

where all access constraints collapse into one cap per branch:

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \underbrace{\bar{h}^O}_{\text{menu}}, \underbrace{\frac{A_i(x) \rho_p}{(1-\phi) q^O}}_{\text{down payment}}, \underbrace{\frac{\psi y_i}{q^O}}_{\text{payment-to-income}} \right\}, \quad \rho_p = r + \delta^O + \tau^P.$$

A renter is capped by the stock; a buyer by whichever of menu, collateral, or debt service is shortest. Policy moves fertility only through the *binding* component of H_i^m .

Note: $P^O = q^O / \rho_p$ enters only the down payment, so a higher property tax *relaxes* the collateral constraint.

Entry and old households

Entry. Logit over inside value vs. outside option $\bar{W}^E$ (taste-shock scale κ_E):

$$\pi_i^E = \frac{\exp\{W_i^Y / \kappa_E\}}{\exp\{W_i^Y / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}.$$

Old incumbents. Endowment H_j^0 ; choose consumption, retained housing, bequests:

$$\max \log c_j^O + \gamma \log h_j^O + \mathcal{B}_j(e_j) \quad \text{s.t.} \quad c_j^O + e_j + (q^O - L_j^\tau)h_j^O = m_j + q^O H_j^0, \quad 0 \leq h_j^O \leq H_j^0,$$

where $L_j^\tau = \tau^P P^O - \tilde{\tau}_j \tilde{P}_j \geq 0$ is the incumbent assessment discount: market tax cost minus protected tax cost.

Releasing a unit earns q^O ; keeping it costs $q^O - L_j^\tau$. The discount is a **subsidy to staying put** — a tax rule, not a preference. Bequests $\mathcal{B}_j$ are a preference and are respected throughout.

Competitive equilibrium

Prices (q^R, q^O) , asset price $P^O = q^O / \rho_p$, construction (I^R, I^O) such that:

- households optimize: branch problems, tenure choice, logit entry; old retention;
- competitive construction: $q^m = \Phi_I^m(I^m; Z^m)$ (flow units);
- both segments clear in aggregate:

$$\bar{M} \int \pi_i^E \mathbf{1}\{o_i=R\} h_i \, dG_Y = \bar{H}^R + I^R, \quad \bar{M} \int \pi_i^E \mathbf{1}\{o_i=O\} h_i \, dG_Y + \int h_j^O \, dF_O = \bar{H}^O + I^O;$$

- housing payments are transfers closed by lump-sum rebates; only construction uses real resources.

Clearing pins aggregates, not assignment: nothing in these conditions asks whether the marginal family-sized unit is held by the household that values it most.

Marginal values and access gaps

A constrained household values space above its price; the gap is the constraint's bite, in goods, read off the bundle:

$$MV_i^m = \frac{\alpha c_i^m}{s_i^m} = q^m + \zeta_i^m, \quad \zeta_i^m = \frac{\alpha c_i^m}{s_i^m} - q^m \geq 0.$$

For owners the gap decomposes by the constraint that generates it:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P^O}_{\text{collateral}} + \underbrace{\frac{\eta_i}{\lambda_i^O}q^O}_{\text{debt service}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{menu}} = \zeta_i^{O,F} + \text{menu part.}$$

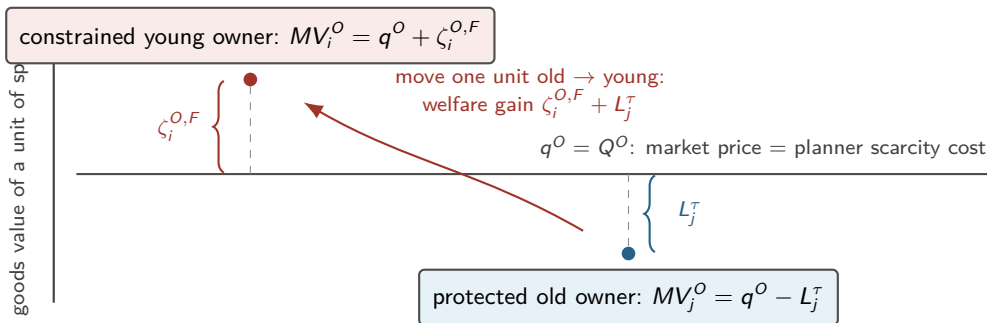
Proposition 1 (fertility condition)

$$\frac{\beta c_i^m}{n_i^m} = \chi_i + \kappa(q^m + \zeta_i^m)$$

A child's space is priced at the household's *own* shadow value of space: children get more expensive exactly where family-sized space is hard to reach, at unchanged market prices.

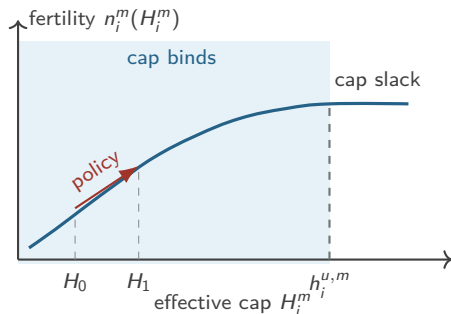
Two gaps separate the equilibrium from the planner

Planner: same preferences, same menus, same construction technology; financial (DP/PTI) constraints relaxable at zero marginal resource cost; fertility-neutral; entry and tenure assignments held at equilibrium.



The equilibrium satisfies the planner's marginal conditions **only if** $\zeta_i^{O,F} = 0$ and $L_j^T = 0$. Binding size *menus* are not a gap (the planner faces the same stock); old occupancy per se is not a gap (true utility and bequests are counted).

Fertility and the family-housing cap



With a binding cap, $h = H$ and

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}}{\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}}.$$

- More space lowers the crowding cost of a child; but the household also buys more housing, squeezing consumption.
- Fertility rises at every binding cap iff $\chi \leq \kappa q^m / \alpha$: children no more goods-intensive than their parents' bundle (one curve per branch).

Equal scaling: the access gap predicts the fertility response

Define the within-branch response $\Psi_i^m = \partial n_i^m / \partial H_i^m$ at a binding cap. Under equal scaling,

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi_i^2}{(c_i^m)^2} + \frac{\alpha \kappa^2}{(s_i^m)^2}} > 0.$$

- $\Psi_i^m > 0$ iff $\zeta_i^m > 0$: fertility responds exactly for the households whose constraint bites, more where it bites harder.
- The entry margin runs on the same object: relaxing the cap raises inside value by $\Theta_i^m = \lambda_i^m \zeta_i^m = \zeta_i^m / c_i^m$ per unit.
- The gap that measures misallocation is the gap that predicts births.

A local policy decomposition

Let $P_i^m(z) = \partial H_i^m / \partial z$ (pass-through to the cap) and $\mathcal{B}_m$ the entrant types whose cap binds in branch m . Holding prices, tenure, and active sets fixed:

$$\left. \frac{dN}{dz} \right|_{q,o,A} = \bar{M} \sum_{m \in \{R,O\}} \int_{\mathcal{B}_m} \left[\underbrace{\pi_i^E \Psi_i^m}_{\text{intensive margin}} + \underbrace{(n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m}_{\text{entry margin}} \right] \underbrace{P_i^m(z)}_{\text{pass-through}} dG_Y(i)$$

- Exposure $\times$ pass-through $\times$ response, plus entry (births net of outside-option fertility $\bar{n}_i^E$).
- The cap is a minimum: a policy that improves a slack component has $P_i^m = 0$ — no effect through this channel. Price-type policies are outside the fixed-price formula.
- Left out: price/construction responses, fiscal costs Ξ^Π (general equilibrium) — and tenure switching, which is first order at fixed prices. Next slide.

Tenure switching at the boundary

Tenure is discrete, so switchers' bundles—and fertility—jump at $W_i^R = W_i^O$. The size of the margin is quantitative; the sign is theory.

Proposition (fertility at a tenure switch)

At indifference between a cheap branch (q^A) and an expensive branch ($q^B > q^A$): the cheap branch's cap binds, and the expensive branch has more housing, more slack space, less consumption. If $\chi \leq \kappa q^A / \alpha$, it also has higher fertility: $n_B > n_A$.

The only reason to pay more per unit of space is to get more space; space-intensive children come with it. Three rent/own cases:

- $q^O > q^R$: switchers into ownership upsize, $n_O > n_R$ — switching term ≥ 0 ; the fixed-tenure formula is a **lower bound** for owner-access policies.
- $q^O = q^R$: allocations continuous at the boundary, no jump — the formula is **exact**.
- $q^O < q^R$: switchers downsize into cheaper owned units, $n_O < n_R$ — switching term ≤ 0 ; an **upper bound**. Which case applies is one benchmark number: owner user cost vs. rent per unit.

Which policies move fertility through this channel

Pass (when the named component binds)

- Down-payment grants, LTV relief (A_i, ϕ):
 $P_i^O = \rho_p / [(1 - \phi)q^O]$ when collateral binds.
- Payment-to-income relaxation (ψ):
 $P_i^O = y_i / q^O$ when debt service binds.
- Family-sized rental supply or conversion ($\bar{h}^R$):
 $P_i^R = 1$ for menu-capped renters.
- Assessment reform: shrinks L_j^T , releases old-held space.

No effect through this channel

- Subsidies to units below the binding size: the binding component does not move.
- Price subsidies that leave H_i^m unchanged: real effects, but they require the GE analysis.
- Liquidity to households with slack constraints: pledgeable wealth has no income effect.

Property tax is two-sided: the level relaxes young down payments (lower P^O); assessment *protection* is what creates old lock-in.

Takeaways

1. **Family-housing access is a fertility margin.** Children are priced at the household's own shadow value of space, $\beta c_i/n_i = \chi_i + \kappa(q^m + \zeta_i^m)$; under $\chi_i \leq \kappa q^m/\alpha$, relaxing a binding cap raises fertility.
2. **Two measurable gaps separate equilibrium from the planner:** the young financing gap $\zeta_i^{O,F}$ and the old assessment discount L_j^T . Reallocating one unit old $\rightarrow$ young is worth $\zeta_i^{O,F} + L_j^T$. The young mechanism stands without the old one.
3. **Policy works only through the binding component of H_i^m .** Local effect = exposure $\times$ pass-through $\times$ response + entry; under equal scaling $\Psi_i^m \propto \zeta_i^m$. Tenure switching reinforces owner-access policies when ownership is the expensive family-space segment, and vanishes at price parity.

Appendix: the branch derivative

If $0 < H < h^u$, the cap binds, $h(H) = H$, and

$$c(H) = y - qH - \chi n(H), \quad s(H) = H - \kappa n(H).$$

Fertility solves $F(n, H) = \frac{\beta}{n} - \frac{\chi}{c(H)} - \frac{\alpha\kappa}{s(H)} = 0$, and

$$F_n = -\left[\frac{\beta}{n^2} + \frac{\chi^2}{c^2} + \frac{\alpha\kappa^2}{s^2}\right] < 0, \quad F_H = \frac{\alpha\kappa}{s^2} - \frac{\chi q}{c^2}, \quad \frac{dn}{dH} = -\frac{F_H}{F_n}.$$

Unconstrained benchmark (Cobb–Douglas in (c, s, n) at prices $(1, q, \chi + \kappa q)$):

$$c^u = \frac{y}{1 + \alpha + \beta}, \quad n^u = \frac{\beta y}{(1 + \alpha + \beta)(\chi + \kappa q)}, \quad \frac{qs^u}{c^u} = \alpha.$$

Appendix: equal scaling and the access gap

At a binding cap, $\alpha c_i^m / s_i^m = q^m + \zeta_i^m$. Under equal scaling $\chi_i = \kappa q^m / \alpha$, the numerator of Ψ_i^m factors:

$$\frac{\alpha \kappa}{(s_i^m)^2} - \frac{\chi_i q^m}{(c_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} \right) \left(\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m} \right), \quad \frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} = \frac{\zeta_i^m}{c_i^m},$$

so $\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m$ with $\Omega_i^m > 0$. The envelope value of relaxing the cap is $\Theta_i^m = \partial W_i^m / \partial H_i^m = \lambda_i^m \zeta_i^m$ with $\lambda_i^m = 1 / c_i^m$, and the logit entry response is $\partial \pi_i^E / \partial W_i^Y = \pi_i^E (1 - \pi_i^E) / \kappa_E$.

Appendix: planner conditions and the comparison

With goods multiplier Λ and segment multipliers ΛQ^m , interior planner allocations satisfy, away from menu corners,

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = Q^m, \quad \frac{u_{h,j}^O}{u_{c,j}^O} = Q^O, \quad Q^m = \Phi_I^m(I^m; Z^m), \quad \frac{\beta c_i}{n_i} + \nu_i^g = \chi_i + \kappa Q_i^{o^P}.$$

Endogenous logit entry enters all allocation FOCs through a common factor that cancels from these ratios.

At an interior competitive allocation, clearing and competitive construction give $Q^R = q^R$, $Q^O = q^O$, and competitive marginal values are

$$MV_i^R = q^R + \zeta_i^R, \quad MV_i^O = q^O + \zeta_i^O, \quad MV_j^O = q^O - L_j^\tau.$$

If DP/PTI are relaxable at zero marginal resource cost, the CE satisfies the planner's conditional FOCs only if $\zeta_i^{O,F} = 0$ and $L_j^\tau = 0$; menus common to both problems generate no gap. Entry and tenure-assignment efficiency are not characterized.

Appendix: accounting conventions

- **Flow normalization.** q^m is both the user cost and the marginal resource cost of segment- m services; Φ^m is in flow units, so $q^m = \Phi_I^m$. P^O only converts q^O into the purchase price for down payments. (Under asset-unit costs, $P^O = \Phi_I^O$ and the property tax would also distort construction.)
- **Stocks.** $\int H_j^0 dF_O + H_{\text{pass}}^O = \bar{H}^O$; rental stock held by passive landlords; passive and fiscal income rebated lump-sum.
- **Bequest loop is open (reduced form).** Old bequest expenditure e_j is a current resource use; young exposure B_i is a primitive of G_Y . The estate tax τ^b works only through pledgeable resources $A_i(x)$.
- **Outside fertility.** N counts births among entrants; with outside births $\bar{n}_i^E$, entry-margin terms use $n_i^Y - \bar{n}_i^E$.